

Ryan Cho

Software Engineer | Mobile Development & Identity Systems

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PROFESSIONAL SUMMARY

Software engineer with seven years of experience delivering secure applications across financial services and enterprise SaaS. Proven ownership of centralized authentication, OAuth and OIDC flows, and SDK-driven integrations for mobile and web clients. Partnered with security and platform teams on CIAM onboarding, token lifecycle design, and mobile hardening. Full-stack engineer skilled in Kotlin, Java, REST services, and cloud-native identity patterns with consistent delivery under regulated environments.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Architected and shipped a **centralized authentication module** for investor-facing mobile and web clients, standardizing sign-in across four product surfaces and reducing duplicate identity code by **60%**.
- Integrated **OAuth 2.0 and OIDC token flows** with headless CIAM providers, enabling SDK and REST-based login paths for Android and iOS teams without shared UI dependencies.
- Led **mobile security hardening** initiatives including certificate pinning, secure token storage, and session timeout policies reviewed by internal security and compliance stakeholders.
- Built reusable **Android authentication SDK wrappers** around enterprise identity services, cutting partner onboarding time from weeks to days for third-party integrations.
- Partnered with platform engineering to define **refresh token rotation** and step-up authentication policies supporting **15K+ monthly active users** across production environments.
- Mentored two junior engineers on **Kotlin coroutines**, navigation-safe auth state, and testable identity layers using Mockito and Espresso.
- Drove CI/CD pipeline updates for signed release builds, **Play App Signing**, and automated security scanning in GitHub Actions for mobile release trains.
- Collaborated with product and legal on **privacy-aware onboarding flows**, biometric opt-in, and audit-ready logging for regulated financial workflows.

Technology Analyst, Incoming Technology Analyst, Technology Summer Analyst,

May/2019 - Apr/2022

BNP Paribas

Remote | Seventh Avenue, New York

- Delivered **enterprise login modernization** for internal banking tools, migrating legacy session models to **SSO-backed OIDC** with centralized policy enforcement.
- Supported **CIAM vendor evaluation and onboarding**, documenting integration patterns for mobile, web, and headless API consumers across technology groups.
- Implemented **role-based access control middleware** in Java services protecting sensitive trading and operations data for **200+ internal users**.
- Built proof-of-concept **Android client** integrating corporate identity SDK with custom MFA prompts aligned to bank security standards.
- Automated identity regression suites covering token expiry, consent screens, and federated logout, improving release confidence for **quarterly deployment cycles**.
- Coordinated with cybersecurity on **mobile threat modeling** and secure credential handling for remote workforce applications during large-scale rollout.
- Produced technical design docs for **API gateway authentication**, rate limiting, and service-to-service JWT validation adopted by three downstream teams.
- Reduced authentication-related production incidents by **40%** through structured logging, alerting, and runbooks for identity service failures.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Developed REST-backed **login and registration flows** for a React web application serving restaurant clients, integrating bcrypt hashing and session cookies.
- Implemented **JWT-based API authentication** for mobile companion features, enabling secure data sync between client devices and backend services.
- Wrote unit and integration tests for auth edge cases including expired tokens and invalid credentials, maintaining **85%+ coverage** on identity modules.
- Refactored password reset and email verification workflows, improving successful recovery rates and reducing support tickets tied to **account access**.
- Collaborated with senior engineers on **API contract design** for OAuth-style delegated access between internal microservices.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Built responsive **user authentication pages** with form validation and server-side session management for a nonprofit web platform.
- Integrated third-party **social login providers** via OAuth redirects, simplifying account creation for first-time visitors.
- Hardened public endpoints against **CSRF and XSS patterns** following OWASP guidance under senior developer review.
- Documented deployment and environment configuration for auth secrets, supporting smooth handoff to **foundation maintainers**.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: Kotlin, Java, Android SDK, Jetpack Compose, React, TypeScript, HTML5, CSS3

Backend: Java, Spring Boot, Node.js, REST APIs, GraphQL, OAuth 2.0, OpenID Connect

Databases: PostgreSQL, MongoDB, Redis, SQLite, Room

Cloud & DevOps: AWS, Docker, Kubernetes, GitHub Actions, CI/CD, Terraform

Monitoring & Tools: Firebase, Sentry, Datadog, Android Studio, Charles Proxy, SonarQube

Workflow & PM: Agile, Scrum, Jira, Confluence, Cross-functional stakeholder collaboration